

Appendix

Anonymous

I. PARAMETER CONFIGURATIONS

In this section, we introduce the parameter configurations employed in our *rHHT*.

In our *rHHT*, a train configuration comprising 1 locomotive and 108 wagons with a total mass of 5000 tons is employed for the simulation. The environment is modeled as a 20-kilometer railway line incorporating three distinct speed limit zones. In the implementation of *rHHT*, the planned time T_p is set to 1600 seconds with $K=80$, *i.e.*, a time step of 20 seconds. Our *rHHT* uses the Davis equation to model the basic resistance W_b , *i.e.*, $W_{b,i} = m_i \cdot (A + Bv_i + Cv_i^2)$, where A , B , and C denote the Davis coefficients, respectively. In this work, the coefficients are set to be constant, *i.e.*, $A = 0.0000209$, $B = 0.000108$, and $C = 0.00767$. Moreover, *rHHT* the Proximal Policy Optimization (PPO) to train the controller. In terms of the reward function parameters, we set the safety reward coefficient w_{sa} is set to 2, the goal reward coefficient w_g is set to 10, the energy reward coefficient w_e is set to 0.1, and the smooth reward coefficient w_s is set to 5. *rHHT* employs the Transformer Encoder component to combine the look-ahead data with the raw state. Specifically, the number of sampling points (*i.e.*, the parameter m) is set to 10, and the dimension of the track feature vector f_t (*i.e.*, the parameter dim_{out}) is set to 4.